
UNIT 7 PROBABILISTIC REASONING

Structure	Page Nos.
7.0 Introduction	235
7.1 Objectives	235
7.2 Reasoning with uncertain information	236
7.3 Review of Probability Theory	236
7.4 Introduction to Bayesian Theory	241
7.5 Bayes' Networks	242
7.6 Probabilistic Inference	242
7.7 Basic idea of Inferencing with Bayes Networks	243
7.8 Other Paradigms of Uncertain Reasoning	244
7.9 Dempster-Shafer Theory	244
7.10 Summary	249
7.11 Solutions/ Answers	250
7.12 Further Readings	251

7.0 INTRODUCTION

This unit is dedicated to probability theory and its usage in decision making for various problems. Contrary to the classical decision making of True and False propositions, the probability of the truth value with a certain probability is used for making decisions. The inclusion of such a probabilistic approach is quite relevant since uncertainties are quite obvious in the real world.

As we know, the probability of an event (uncertain event I) is basically the measure of the degree of likelihood of the occurrence of event I. Let the set of all such possible events be represented as the sample space S. The measure of probability is a function $P()$ mapping the event outcome E_i from the sample space S to some real number and satisfying a few conditions, such as:

- (i) $0 \leq P(I) \leq 1$ for any event $I \subseteq S$
- (ii) $P(S) = 1$, represents a certain outcome, and
- (iii) For $E_i \cap E_j = \phi$, for all $i \neq j$ (the E_i are mutually exclusive), i.e. $P(E_1 \cup E_2 \dots) = P(E_1) + P(E_2) + \dots$

Using the above mentioned three conditions, we can derive the basic laws of probability. It should also be noted that only these three conditions are not enough to compute the probability of an outcome. This additionally requires the collection of experimental data for estimating the underlying distribution.

7.1 OBJECTIVES

After going through this unit, you should be able to:

- Understand the role of probabilistic reasoning in AI
- Understand the concept of Bayesian theory and Bayesian networks
- Perform probabilistic inference through Bayesian Networks

- Understand the other Paradigm of Uncertain Reasoning & Dempster-Shafer Theory

7.2 REASONING WITH UNCERTAIN INFORMATION

Reasoning is an important component of various decision-making processes. The amount of information and its correctness play a crucial role in reasoning. Decision-making is easier when we have certain information, i.e., the correctness of the information can be ascertained. In situations when the certainty of information cannot be ascertained, the decision-making process is likely to be erroneous. How decisions are made under uncertainty is the core focus of this unit. Sources of uncertainty in information may arise due to various reasons, including experimental errors, instrument faults, unreliable sources, and other factors. Once information is received, decisions often have to be made based on uncertain information, and we cannot rely on models that use certain information. One potential solution for such scenarios is probabilistic reasoning. Probabilistic models can be used to reason under uncertainty and support decision-making. Let's first see the basic probability concepts before discussing probabilistic reasoning..

7.3 REVIEW OF PROBABILITY THEORY

Now, you are familiar with the reasoning and how it can be useful with probability theory. Before we dive deeper into Bayes' theorem and its applications, let us review some of the basic concepts of probability theory. These concepts will help us to understand other topics in this unit.

Trials, Sample Space, Events: You must have often observed that a random experiment may comprise a series of smaller sub-experiments. These are called trials. Consider, for instance, the following situations.

Example 1: Suppose the experiment consists of observing the results of three successive tosses of a coin. Each toss is a trial, and the experiment consists of three trials, so that it is completed only after the third toss (trial) is over.

Example 2: Suppose from a lot of manufactured items, ten items are chosen successively following a certain mechanism for checking. The underlying experiment is completed only after the selection of the tenth item is completed; the experiment obviously comprises 10 trials.

Example 3: If you consider Example 1 once again, you would notice that each toss (trial) results in either a head (H) or a tail (T). In all there are 8 possible outcomes of the experiment viz., $s_1 = (H, H, H)$, $s_2 = (H, H, T)$, $s_3 = (H, T, H)$, $s_4 = (T, H, H)$, $s_5 = (T, T, H)$, $s_6 = (T, H, T)$, $s_7 = (H, T, T)$ and $s_8 = (T, T, T)$.

Let ζ be a fixed sample space. We have already defined an event as a collection of sample points from ζ . Imagine that the (conceptual) experiment underlying ζ is being performed. The phrase "the event E occurs" would mean that the experiment results in an outcome that is included in the event E. Similarly, non-occurrence of the event E would mean that the experiment results into an outcome that is not an element of the event E. Thus, the collection of all sample points that are not included in the event E is also an event which is complementary to E and is denoted as E^c . The event E^c is therefore the event that contains all those sample points of ζ which are not in E. As such, it is easy

to see that the event E occurs if and only if the event E^c does not take place. The events E and E^c are complementary events and taken together they comprise the entire sample space, i.e., $E \cup E^c = \zeta$.

You may recall that ζ is an event that consists of all the sample points. Hence, its complement is an empty set in the sense that it does not contain any sample point and is called the null event, usually denoted as $\emptyset$, so that $\zeta^c = \emptyset$.

Let us once again consider Example 3. Consider the event E that the three tosses produce at least one head. Thus, $E = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}$ so that the complementary event $E^c = \{s_8\}$, which is the event of not scoring a head at all. Again in Example 3 in the case of selection without replacement, event that the white marble is picked up at least once is defined as $E = \{(r_1, w), (r_2, w), (w, r_2), (w, r_1)\}$. Hence, $E^c = \{(r_1, r_2), (r_2, r_1)\}$ i.e. the event of not picking the white marble at all.

Let us now consider two events E and F . We write $E \cup F$, read as E "union" F , to denote the collection of sample points, which are responsible for the occurrence of either E or F or both. Thus, $E \cup F$ is a new event and it occurs if and only if either E or F or both occur i.e. if and only if at least one of the events E or F occurs. Generalizing this idea, we can define a new event E_j , read as "union" of the k events $E_1, E_2, \dots, E_k$, as the event which consists of all sample points that are in at least one of the events $E_1, E_2, \dots, E_k$ and it occurs if and only if at least one of the events $E_1, E_2, \dots, E_k$ occurs.

Again, let E and F be two given events. We write $E \cap F$, read as E "Intersection" F , to denote the collection of sample points any of whose occurrence implies the occurrence of both E and F . Thus, $E \cap F$ is a new event, and it occurs if and only if both the events E and F occur. Generalizing this idea, we can define a new event E_j read as "intersection" of the k events $E_1, E_2, \dots, E_k$, as the event which consists of sample points that are common to each of the events $E_1, E_2, \dots, E_k$, and it occurs only if all the k events $E_1, E_2, \dots, E_k$ occur simultaneously. Further, two events E and F are said to be mutually exclusive or disjoint if they do not have a common sample point i.e. $E \cap F = \emptyset$.

Two mutually exclusive events cannot occur simultaneously. In the coin-tossing experiment, for instance, the two events, heads and tails, are mutually exclusive: if one occurs, the other cannot occur. To have a better understanding of these events, let us once again look at Example 3. Let E be the event of scoring an odd number of heads and F be the event that a tail appears in the first two tosses, so that $E = \{s_1, s_5, s_6, s_7\}$ and $F = \{s_5, s_8\}$. Now $E \cap F = \{s_5\}$, the event that only the third toss yields a head. Thus, events E and F are not mutually exclusive.

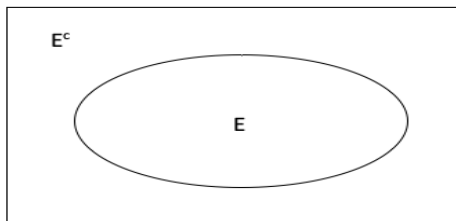


Fig. 1(a)

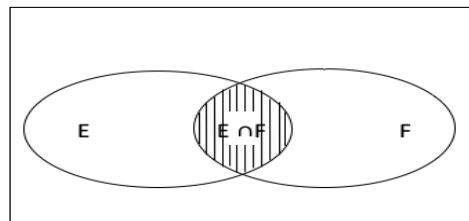


Fig.1(b)

The above relations between events can be best viewed through a Venn diagram. A rectangle is drawn to represent the sample space ζ . All the sample

points are represented within the rectangle by means of points. An event is represented by the region enclosed by a closed curve containing all the sample points leading to that event. The space inside the rectangle but outside the closed curve representing E represents the complementary event E^c (See Fig.1(a) above). Similarly, in Fig.1(b), the space inside the curve represented by the broken line represents the event $E \cup F$, and the shaded portion represents $E \cap F$.

As is clear by now, because the outcome of a random experiment is uncertain, none of the various events associated with a sample space can be predicted with certainty before the underlying experiment is performed and the outcome of it is noted. However, some events may intuitively seem to be more likely than the rest. For example, in the context of human beings, the event that a person will live 20 years seems to be more likely compared to the event that the person will live 200 years. Such thoughts motivate us to explore if one can construct a scale of measurement to distinguish between the likelihoods of various events. Towards this, a small but extremely significant fact comes to our help. Before we elaborate on this, we need a couple of definitions.

Consider an event E associated with a random experiment; suppose the experiment is repeated n times under identical conditions and suppose the event E (which is not likely to occur with every performance of the experiment) occurs $f_n(E)$ times in these n repetitions. Then, $f_n(E)$ is called the frequency of the event E in n repetitions of the experiment and $r_n(E) = f_n(E)/n$ is called the relative frequency of the event E in n repetitions of the experiment. Let us consider the following example.

Example 4: Consider the experiment of throwing a coin. Suppose we repeat the process of throwing a coin 5 times and suppose the frequencies of occurrence of heads are tabulated below in Table-1:

Table 1:

No. of repetitions (n)	Frequency of head ($f_n(H)$)	Relative frequency of head $r_n(H)$
1	0	0
2	1	1/2
3	2	2/3
4	3	3/4
5	3	3/5

Notice that the third column in Table-1 gives the relative frequencies $r_n(H)$ of heads. We can keep on increasing the number of repetitions n and continue calculating the values of $r_n(H)$ in Table 1. Merely to fix ideas regarding the concept of probability of an event, we present below a very naive approach which in no way is rigorous, but it helps to see things better at this stage.

Check Your Progress- 1

1. In each of the following exercises, an experiment is described. Specify the relevant sample spaces:

- a) A machine manufactures a certain item. An item produced by the machine is tested to determine whether or not it is defective.
- b) An urn contains six balls, which are colored differently. A ball is drawn from the urn and its color is noted.
- c) An urn contains ten cards numbered 1 through 10. A card is drawn, its number noted and the card is replaced. Another card is drawn and its number is noted.

.....
.....
.....

2. Suppose a six-faced die is thrown twice. Describe each of the following events:

- i) The maximum score is 6.
- ii) The total score is 9.
- iii) Each throw results in an even score.
- iv) Each throw results in an even score larger than 2.
- v) The scores on the two throws differ by at least 2.

.....
.....
.....

7.3.1 Conditional probability and independent events

Let ζ be the sample space corresponding to an experiment and let E and F be two events of ζ . Suppose the experiment is performed and it is only known that the event F has taken place. Thus, there still remains a scope for speculation about the occurrence of the other event E . In view of this additional information confirming the occurrence of F in view, it would be appropriate to modify the probability of occurrence of E suitably. That such modifications would be necessary can be readily appreciated through two simple instances as follows:

Example 5: Suppose E and F are such that $F \subseteq E$, so that the occurrence of F would automatically imply the occurrence of E . Thus, with the information that the event F has taken place in view, it is plausible to assign probability 1 to the occurrence of E irrespective of its original probability.

Example 6: Suppose E and F are two mutually exclusive events, and thus they cannot occur together. Thus, whenever we come to know that the event F has taken place, we can rule out the occurrence of E . Therefore, in such a situation, it will be appropriate to assign probability 0 to the occurrence of E .

Example 7: Suppose a pair of balanced dice A and B is rolled simultaneously so that each of the 36 possible outcomes is equally likely to occur and hence has probability $\frac{1}{36}$. Let E be the event that the sum of the two scores is 10 or more, and F be the event that exactly one of the two scores is 5.

Then $E = \{(4, 6), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$ so that $P(E) = 6/36 = 1/6$.

Also, $F = \{(1, 5), (2, 5), (3, 5), (4, 5), (6, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 6)\}$.

Now, suppose we are told that the event F has taken place (note that this is only partial information relating to the outcome of the experiment). Since each of the outcomes originally had the same probability of occurring, they should still have equal probabilities. Thus, given that exactly one of the two scores is 5, each of the 10 outcomes of event F has a probability, while the probability of the remaining 26 points in the sample space is 0.

In light of the information that the event F has taken place, the sample points $(4, 6)$, $(6, 4)$, $(5, 5)$, and $(6, 6)$ in the event E must not have materialized. One of the two sample points $(5, 6)$ or $(6, 5)$ must have materialized. Therefore, the probability of E would no longer be $1/6$. Since all the 10 sample points in F are equally likely, the revised probability of E given the occurrence of F , which occurs through the materialization of one of the two sample points $(6, 5)$ or $(5, 6)$, should be $2/10 = 1/5$.

The probability just obtained is called the conditional probability that E occurs, given that F has occurred and is denoted by $P(E|F)$. We shall now derive a general formula for calculating $P(E|F)$.

Consider the following probability table:

Table 2

Events	E	E^c
F	p	q
F^c	r	s

In Table 2, $P(E \cap F) = p$, $P(E^c \cap F) = q$, $P(E \cap F^c) = r$ and $P(E^c \cap F^c) = s$ and hence, $P(E) = P(E \cap F) \cup (E \cap F^c) = P(E \cap F) + P(E \cap F^c) = p + r$ and similarly, $P(F) = q + s$.

Now suppose that the underlying random experiment is being repeated a large number of times, say N times. Thus, taking a cue from the long-term relative frequency interpretation of probability, the approximate number of times the event F is expected to take place will be $NP(F) = N(q+s)$. Under the condition that the event F has taken place, the number of times the event E is expected to take place would be $NP(E \cap F)$, as both E and F must occur simultaneously. Thus, the long-term relative frequency of E under the condition of occurrence of F , i.e., the probability of occurrence of E under the condition of occurrence of F , should be $NP(E \cap F)/NP(F) = P(E \cap F)/P(F)$. This is the proportion of times E occurs out of the repetitions where F takes place. With the above background, we are now ready to formally define the conditional probability of an event given another.

Definition: Let E and F be two events from a sample space ζ . The conditional probability of the event E given the event F , denoted by $P(E|F)$, is defined as $P(E|F) = P(E \cap F)/P(F)$, whenever $P(F) > 0$.

When $P(F) = 0$, we say that $P(E|F)$ is undefined. We can also write from Eqn. $P(E \cap F) = P(E|F)P(F)$.

Referring back to Example 3, we see that $P(E) = 6/36$, $P(F) = 10/36$; since, $E \cap F = \{(5,6), (6,5)\}$, $P(E \cap F) = 2/36$, $P(E|F) = (2/36)/(10/36) = 2/10 = 1/5$, which is the same as that obtained in Example 3. Another result can be generalized to k events $E_1, E_2, \dots, E_k$, where $k \geq 2$. And now an exercise for you.

Check Your Progress 2

1. In a class, three students tossed one coin (one each) for 3 times. Write down all the possible outcomes that can be obtained in this experiment.

.....

2. In problem 1, what is the probability of getting 2 more than 2 heads at a time? Also, write the probability of getting three tails at a time.

.....

3. In problem 1, calculate the Relative frequency of tail $r_n(T)$.

.....

7.4 INTRODUCTION TO BAYESIAN THEORY

Bayes' theorem is widely used to calculate the conditional probabilities of events without knowing the joint probability. It is also used to calculate the conditional probability where intuition fails. In simple terms, the conditional probability of a hypothesis H given E can be defined as $P_E(H) = P(H \& E) / P(E)$, where $P(E) > 0$, and the term $P(H \& E)$ also exists. Here, P_E is referred to as a probability function. To simply understand Bayes' theorem, have a look at the following definitions.

Joint Probability: This refers to the probability of two or more events simultaneously occurring, e.g. $P(A \text{ and } B)$ or $P(A, B)$.

Marginal Probability: It is the probability of an event occurring irrespective of the outcome of the other random variables, e.g., $P(A)$.

Conditional probability: A conditional probability is defined as the probability of the occurrence of an event provided that another event has occurred. e.g. $P(A | B)$.

The conditional probability can also be written in terms of joint probability as $P(A|B) = P(A, B) / P(B)$. In another way, if one conditional probability is given, the other can be calculated as $P(A|B) = P(B|A) * P(A) / P(B)$.

Let 'S' be a sample space under consideration. Let events ' A_1 ', ' A_2 ', $\dots$, ' A_n ' be a set of mutually exclusive events in the sample space 'S'. Let 'B' be an event from sample space 'S' provided $P(B) > 0$, then according to Bayes' theorem.

$P(A_k | B) = P(A_k \cap B) / P(A_1 \cap B) + P(A_2 \cap B) \dots \dots \dots P(A_n \cap B)$, this can also be written in terms of Bayes' theorem.

$$P(A_k | B) = P(A_k)P(B | A_k) / P(A_1).P(B | A_1)+P(A_2).P(B | A_2) \dots \dots \dots .P(A_n).P(B | A_n)$$

7.5 BAYE'S NETWORKS

The probabilistic models are being used in defining the relationships among variables and are used to calculate probabilities. The Bayes' network is a simpler form of applying Bayes' theorem to complex real-world problems. It uses a probabilistic graphical model that explicitly captures conditional dependencies through directed edges in a graph. Here, if we take fully conditional models, we may need a big amount of data to address all possible events/ cases, and in such a scenario, probabilities may not be calculated practically. On the other hand, simple assumptions like conditional independence of random variables may turn out to be effective, giving a way for Bayes' Network.

While representing a Bayes' Network graphically, nodes represent the distribution of probabilities for random variables. The edges in the graph represent the relationship among random variables. The key benefits of a Bayes' Network are model visualization, relationships among random variables, and computations of complex probabilities.

Example 8: Let us now create a Bayesian Network for an example problem. Let us consider three random variables A, B, and C. It is given that A is dependent on B, and C is dependent on B. The conditional dependence can be stated as $P(A|B)$ and $P(C|B)$ for both the given statements, respectively. Similarly, the conditional independence can be stated here as $P(A|B, C)$ and $P(C|B, A)$ for both statements, respectively.

Here, we can also write $P(A|C, B) = P(A|B)$ as A is unaffected by C. Also, the joint probability of A and C given B can be written as a product of conditional probabilities as $P(A, C|B) = P(A|B) * P(C|B)$.

Now using Bayes' theorem, the joint probability of $P(A, B, C)$ can be written as $P(A, B, C) = P(A|B) * P(C|B) * P(B)$.

The corresponding graph is shown below in Figure 2. Here, each random variable is represented as a node, and edges between nodes are conditional probabilities.

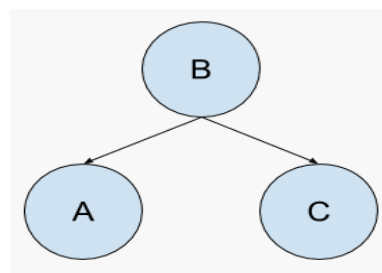


Fig. 2

7.6 PROBABILISTIC INFERENCE

The probabilistic inference is very much dependent on the conditional probability of the specified events, provided the information on the occurrence of other events is available. For two events E and F such that $P(F) > 0$, the conditional probability of event E when F has occurred can be written as:

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

When an experiment is repeated a large number of times (say n) the above expression can be given a frequency interpretation. Let the number of occurrences of an event F be represented as $No. (F)$ and the probability of a joint event of E and F as $No. (E \cap F)$. The relative frequencies of both these events can be computed as f_r :

$$f_r(E \cap F) = \frac{No.(E \cap F)}{n} \quad \text{and similarly,} \quad f_r(F) = \frac{No.(F)}{n}$$

Here, if n is large, the ratio of the above two expressions represents the proportion of times the event E occurs relative to the occurrence of F . This can also be interpreted as an approximation of the conditional probability of E given F .

$$f_r(E \cap F) / f_r(F) \simeq P(E \cap F) / P(F)$$

We can also write the conditional probability of event F given that event E has already occurred, as

$$P(E / F) = P(E \cap F) / P(F)$$

Using the above two equations, we can also write

$$P(F / E) = P(E / F) P(F) / P(E)$$

The above expression is also one form of Bayes' Rule. Here, the notion is simple: the probability of an event F occurring when we know the probability of an event E which has already occurred is the same as the probability of occurring of event E when the probability of occurrence of event F is known.

7.7 BASIC IDEA OF INFERENCING WITH BAYE'S NETWORKS

We are now aware of the Bayes theorem, probability, and Bayes networks. Let's now talk about how inferences can be made using Bayes networks. A network represents the degree of belief of proposition and their causal interdependence. The inference in a network can be done by propagating the given probabilities of related information through the network, producing output at one of the conclusion nodes. The network representation also reduces the time and space requirements for huge computations involving the probabilities of uncertain knowledge of propositional variables. Furthermore, making inferences from such large volumes of data in real time may be impractical. A solution to such a problem can be found using network representation. Here, the network of nodes represents variables connected by edges, which represent causal influences (dependencies) among nodes. Here, the edge weights can be used to represent the strength of influences, or in other terms, the conditional probabilities.

To use this type of probabilistic inference model, one first needs to assign probabilities to all basic facts in the underlying knowledge base. This requires the definition of an appropriate sample space and the assignment of a priori and conditional probabilities. In addition to this, some methods must be selected to compute the combined probabilities when pooling evidence in a sequence of inference steps. In the end, when the outcome of an inference chain results in one or more proposed conclusions, the alternatives must be compared and one should be chosen on the basis of likelihood.

7.8 OTHER PARADIGMS OF UNCERTAIN REASONING

Other approaches to dealing with uncertainty do not have a formal theoretical foundation. These are mostly based on intuition. These approaches are selected over formal methods as a pragmatic solution to a particular problem when the formal methods impose difficult or impossible conditions. One such ad hoc procedure is used to diagnose meningitis and infectious blood disease; this system is called MYCIN. The MYCIN uses If and then rules to assess various forms of patient evidence. It also measures both belief and disbelief to represent the degree of confirmation and disconfirmation, respectively, in a given hypothesis. The ad hoc methods have been used in a larger number of knowledge-based systems than formal methods. This is due to the difficulties encountered in acquiring a large number of reliable probabilities related to the given domain and to the complexities to the ensuing calculations.

One other paradigm is to use Heuristic reasoning methods. These are based on the use of procedures, rules and other forms of encoded knowledge to achieve specified goals under certainty. Using both domain specific and general heuristics, one of several alternative conclusions may be chosen through the strength of positive vs negative evidence presented in the form of justification or endorsement.

The in depth and detailed discussion on this is not in the scope of this unit/course.

7.9 DEMPSTER-SHAFER THEORY

Let us now discuss a mathematical theory based only on evidence, known as Dempster-Shafer (D-S) theory, given by Dempster and extended by Shafer in “Mathematical Theory of Evidence”. This uses a belief function to combine separate and independent pieces of evidence to quantify the belief in a statement. The D-S theory is a generalization of Bayesian probability theory where multiple possible events are assigned probabilities, as opposed to mutually exclusive singletons. The D-S theory assumes that incomplete knowledge creates uncertainty, which in turn gives rise to belief. Here, the uncertainty of the hypothesis is represented by the belief function. The main characteristics of the theory are:

1. Probabilities may be assigned to multiple possible events.
2. These events should be exhaustive and mutually exclusive.

Here, the multiple sources of information are assigned some degree of belief and then aggregated using the D-S combination rule. However, this limits the theory in computationally intensive applications because independent assumptions are often difficult to make when dealing with a large number of information sources.

Let us now define a few terms used in D-S theory that will be useful to us.

7.9.1 Evidence

These are events related to a hypothesis or a set of hypotheses. Here, a relationship is not assumed between various pieces of evidence or among a set of hypotheses. Also, the relationship between a set of hypotheses and a piece of evidence is quantified using data from a source. In the context of D-S theory, we have four types of evidence as follows:

- a) Consonant Evidence: These typically appear in a nested structure where each subset is contained within the next larger subset and so on. As the subset size increases, the information progressively refines the evidential set.
- b) Consistent Evidence: This implies the presence of at least one common element to all the subsets.
- c) Arbitrary Evidence: A situation in which there is no common element occurring in all the subsets, although some subsets may share common elements.
- d) Disjoint Evidence: No subset has common elements with another subset.

All four types of evidence can be understood with the help of Figure 3(a–d) given below.

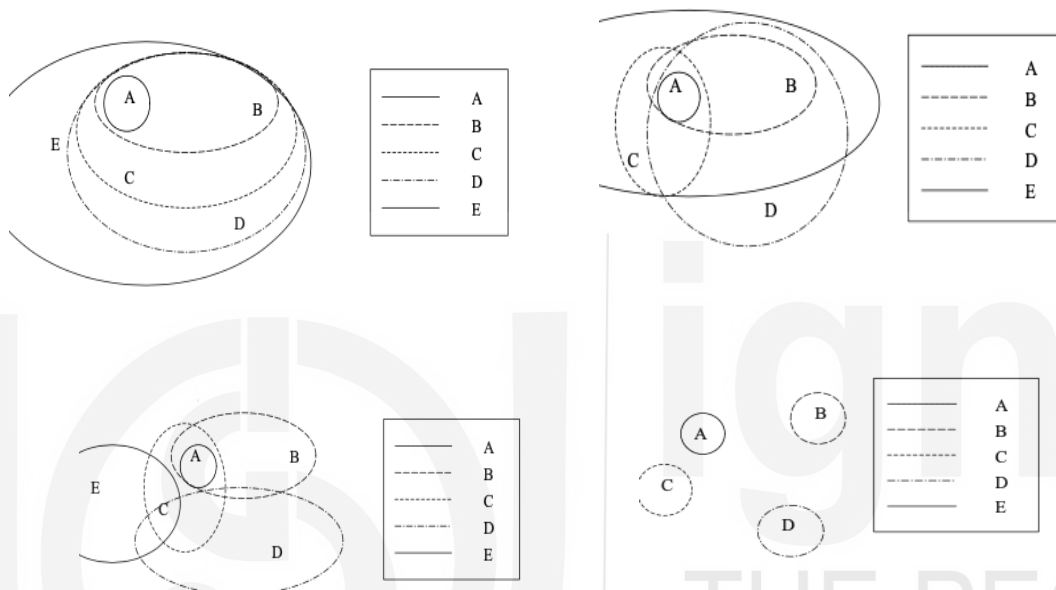


Figure 3. (a-d)

A source of information can be an entity or a person providing relevant state information. Here, the information source is assumed to be non-biased. Information received from such sources is combined to produce more reliable information for further use. D-S theory models are able to handle varying levels of precision in information, and hence, no additional assumptions are required to represent such information.

7.9.2 Frame of Discernment

Let us consider a random variable ' θ ' whose true value is not known. Let ' θ ' = $\{ \theta_1, \theta_2, \dots, \theta_n \}$ represent the mutually exclusive and discrete values of the possible outcomes of ' θ '. Conventionally, the uncertainty about ' θ ' is given by the assigning probability p_i to the elements θ_i , $i = 1: n$, satisfying $\sum p_i = 1$. In the case of D-S theory, the probabilities are assigned to the subsets of ' θ ' as well as to the individual element ' θ_i '.

7.9.3 The Power Set $P(\theta) = 2^{\{\theta\}}$

This is defined as the set of all subsets of ' θ ' including singletons, defining the frame of ' θ '. The subset of this powerset may contain a single or a conjunction of hypotheses. Here, with respect to the power set, the complete probability assignment is called basic probability assignment.

The core functions in D-S theory are:

1. Basic Probability Assignment function

This is represented by m and maps the power set to the interval 0 and 1. Here, the basic probability assignment (bpa) to the null set is 0 and for all subsets of the power set $\sum = 1$. For a given set A , $m(A)$ represents the measure of belief assigned by the available evidence in support of A , where $A \in 2^\theta$. Mathematically, the bpa can be represented as follows.

1. $m : 2^\theta \rightarrow [0, 1]$ (interval)
2. $m(\phi) = 0$ (null)
3. $m(A) \geq 0, \forall A \in 2^\theta$
4. $\sum\{m(A) \mid \forall A \in 2^\theta\} = 1$.

This is to note here that the element of power set with $m(A) > 0$ is termed as a focal element(s).

Example 9: Let $\theta = \{a, b, c\}$; then the power set is $P(\theta) = \{\phi, a, b, c, (a, b), (a, c), (b, c), (a, b, c)\}$. The information source assigned the m -values as $m(a) = 0.2$, $m(c) = 0.1$ and $m(a, b) = 0.4$. Here, the mentioned three subsets are focal elements.

2. The Belief Function

The assignment of basic probabilities allows us to define the lower and upper bounds of the intervals representing the probability of a set. This is also bounded by continuous non-additive measures known as Belief and Plausibility.

The lower bound (belief) for set A is defined as the sum of all basic probability assignments of a proper subset B of set A . The belief function measures the amount of support provided by the information source for a specific element, as a correct one is done by the belief function, mathematically $\text{Bel}(A) = \sum_{\{B \subset A\}} m(B) \quad \forall A \subset \theta$.

3. The Plausibility Function

The upper bound (plausibility) for set A is defined as the sum of all basic probability assignments of B intersecting set A , mathematically $\text{Pl}(A) = \sum_{\{B \cap A \neq \phi\}} m(B)$. Here, the plausibility function measures the level of information by a source contradicting an element being the correct answer, specifically.

Apart from the above-mentioned functions, a few additional terms require attention when discussing the D-S theory. The Uncertainty Interval shows the range within which the true probability may be

found. This is calculated as the difference of belief and plausibility level i.e., $\text{Pl}(A) - \text{Bel}(A)$.

7.9.4 Rule of Combination

In the D-S theory, the measure of Plausibility and Belief is taken from the combined assignments. The D-S rule of combination combines multiple belief functions using their respective basic probability assignments (m). The D-S combination rule is basically a conjunctive operation, i.e., AND. Here, the joint

$m_{\{12\}}$ (combination) is obtained by aggregating the two basic probability assignments m_1 and m_2 as follows:

$$m_{\{12\}}(A) = 1 / \sum_{\{B \cap C = A\}} m_1(B) m_2(C) \{1-K\},$$

Where, $A \neq \phi$,

$$M_{\{12\}}(\phi) = 0,$$

$$\text{And } K = \sum_{\{B \cap C = \phi\}} m_1(B) m_2(C).$$

Here, in the above expression, K is the basic probability mass associated with the conflict and is calculated as the sum of the products of the basic probability assignments of all sets having null intersection. The normalization factor is represented as $1-K$ in the denominator. The rule is associative, commutative, but not continuous or idempotent in nature.

Example 10: In a multinational company, 100 applicants appeared for a job interview. The company set up two interview boards for applicants.

While assessing the grades of the class of 100 students, two of the class teachers responded to the overall result as follows. The first teacher assessed that 40 students will get A grade and 20 students will get a B grade amongst the total 60 students he interviewed. Whereas the second teacher stated that 30 students will get A grade and 30 students will get either A or B grade, amongst the 60 students he took the interview. Combining both evidences to find the resultant evidence, we will do the following calculations. Here, the frame of discernment $\theta = \{A, B\}$ and the power set $2^\theta = \{\emptyset, A, B, (A, B)\}$,

Evidence (1) = Ev1

$$m_1(A) = 0.4$$

$$m_1(B) = 0.2$$

$$m_1(\theta) = 0.4$$

Evidence (2) = Ev2

$$m_2(A) = 0.3$$

$$m_2(A, B) = 0.3$$

$$m_2(\theta) = 0.4$$

Plausibility function (PI):

$$A \cap A = A \neq \emptyset \text{ hence } m_1(A) = 0.4$$

$$A \cap B = \emptyset$$

$$A \cap \theta = A \neq \emptyset \text{ hence } m_1(\theta) = 0.4$$

$$Pl_1(A) = m_1(A) + m_1(\theta) = 0.4 + 0.4 = 0.8$$

$$A \cap A = A \neq \emptyset \text{ hence } m_2(A) = 0.3$$

$$A \cap B = \emptyset$$

$$A \cap \theta = A \neq \emptyset \text{ hence } m_2(\theta) = 0.4$$

$$Pl_2(A) = m_2(A) + m_2(\theta) = 0.3 + 0.4 = 0.7$$

$$B \cap A = \emptyset$$

$$B \cap B = B \neq \emptyset \text{ hence } m_1(B) = 0.2$$

$$B \cap \theta = B \neq \emptyset \text{ hence } m_1(\theta) = 0.4$$

$$Pl_1(B) = m_1(B) + m_1(\theta) = 0.2 + 0.4 = 0.6$$

$$(A, B) \cap A = A \neq \emptyset \text{ } m_2(A) = 0.3$$

$$(A, B) \cap B = B \neq \emptyset, m_2(B) = 0$$

$$(A, B) \cap (A, B) = (A, B) \neq \emptyset \text{ } m_2(A, B) = 0.3$$

$$(A, B) \cap \theta = (A, B) \neq \emptyset \text{ hence } m_2(\theta) = 0.4$$

$$Pl_1(A, B) = m_2(A) + m_2(A, B) + m_2(\theta) = 0.3 + 0.3 + 0.4 = 1.0$$

$$\theta \cap A = A \neq \emptyset \text{ hence } m_1(A) = 0.4$$

$$\theta \cap B = B \neq \emptyset \text{ hence } m_1(B) = 0.2$$

$$\theta \cap \theta = \theta \neq \emptyset \text{ hence } m_1(\theta) = 0.4$$

$$Pl_1(\theta) = m_1(A) + m_1(B) + m_1(\theta) = 0.4 + 0.2 + 0.4 = 1.0$$

$$\theta \cap A = A \neq \emptyset \text{ hence } m_2(A) = 0.3$$

$$\theta \cap (A, B) = (A, B) \neq \emptyset, m_2(A, B) = 0.3$$

$$\theta \cap \theta = \theta \neq \emptyset \text{ hence } m_2(\theta) = 0.4$$

$$Pl_2(\theta) = m_2(A) + m_2(A, B) + m_2(\theta) = 0.3 + 0.3 + 0.4 = 1.0$$

D-S Rule of Combination: Table 3 shows the combination of concordant evidence using D-S theory.

Evidences	m1(A)=0.4		m1(B)=0.2		m1(θ)=0.4
m2(A)=0.3	m1-2 (A)	0.12	m1-2 (∅)	0.06	m1-2 (A) 0.12
m2(A, B)= 0.3	m1-2 (A)	0.12	m1-2 (B)	0.06	m1-2 (A,B) 0.12

$m_2(\theta)=0.4$	$m_{1-2}(A)$	0.16	$m_{1-2}(B)$	0.08	$m_{1-2}(\theta)$	0.16
-------------------	--------------	------	--------------	------	-------------------	------

$k = 0.06$ and $1 - k = 0.94$ Combined masses are worked out

$$\text{Bel}_{1-2}(A) = m_{1-2}(A) = 0.553$$

$$\text{Bel}_{1-2}(B) = m_{1-2}(B) = 0.149$$

$$\text{Bel}_{1-2}(A, B) = m_{1-2}(A) + m_{1-2}(B) + m_{1-2}(A, B) = 0.553 + 0.149 + 0.128 =$$

$$0.83 \quad \text{Bel}_{1-2}(\theta) = m_{1-2}(A) + m_{1-2}(B) + m_{1-2}(A, B) + m_{1-2}(\theta) = 0.553 +$$

$$0.149 + 0.128 +$$

$$0.170 = 1$$

$$\text{Pl}_{1-2}(A) = m_{1-2}(A) + m_{1-2}(A, B) + m_{1-2}(\theta) = 0.553 + 0.128 + 0.170 = 0.851,$$

$$(85 \text{ students in A Grade})$$

$$\text{Pl}_{1-2}(B) = m_{1-2}(B) + m_{1-2}(A, B) + m_{1-2}(\theta) = 0.149 + 0.128 + 0.170 = 0.447,$$

$$(45 \text{ students in B Grade})$$

$$\text{Pl}_{1-2}(A, B) = m_{1-2}(A) + m_{1-2}(B) + m_{1-2}(AB) + m_{1-2}(\theta) = 0.553 + 0.149 +$$

$$0.128 + 0.170 = 1.0$$

$$\text{Pl}_{1-2}(\theta) = m_{1-2}(A) + m_{1-2}(B) + m_{1-2}(A, B) + m_{1-2}(\theta) = 0.553 + 0.149 + 0.128 + 0.170 =$$

$$1.00. (100 \text{ students in total})$$

According to the rule of combination, the concluded ranges are then 55 to 85 students will get

“A” grade and 15 to 45 students will get “B” grade.

Key advantages of the D-S theory:

- The level of uncertainty reduces with the addition of information.
- Addition of more evidence reduces ignorance
- We can represent diagnostic hierarchies using D-S theory.

Check Your Progress 3

Problem-1. Differentiate between joint, marginal, and conditional probability with an example of each.

Problem-2. Explain the Dempster-Shafer theory with a suitable example.

Problem-3. What are the different type of evidence? Give a suitable example of each.

7.10 SUMMARY

This unit relates to the discussion of reasoning under uncertainty, which involves a review of probability theory and an introduction to Bayesian Theory. The unit also

covers the concept of Bayes' Networks, which is later used for inference. Finally, the unit also discusses other paradigms of uncertain reasoning, including the Dempster-Shafer Theory.

7.11 SOLUTIONS/ANSWERS

Check Your Progress- 1

Problem -1. In each of the following exercises, an experiment is described. Specify the relevant sample spaces:

- a) A machine manufactures a certain item. An item produced by the machine is tested to determine whether or not it is defective.
- b) An urn contains six balls, which are colored differently. A ball is drawn from the urn and its color is noted.
- c) An urn contains ten cards numbered 1 through 10. A card is drawn, its number noted and the card is replaced. Another card is drawn and its number is noted.

Solution - *Please refer to section 7.3 to answer these problems.

Problem 2. Suppose a six-faced die is thrown twice. Describe each of the following events:

- i) The maximum score is 6.
- ii) The total score is 9.
- iii) Each throw results in an even score.
- iv) Each throw results in an even score larger than 2.
- v) The scores on the two throws differ by at least 2.

Solution - *Please refer to section 7.3 to answer these problems.

Check Your Progress 2

Problem-1: In a class, three students tossed one coin (one each) for 3 times. Write down all the possible outcomes which can be obtained in this experiment.

Solution - *Please refer to example 4 and section 7.3 to solve these problems

Problem-2: In problem 1, what is the probability of getting 2 more than 2 heads at a time. Also write the probability of getting three tails at a time.

Solution - *Please refer to example 4 and section 7.3 to solve these problems

Problem-3: In problem 1 calculate the Relative frequency of tail $r_n(T)$.

Solution - *Please refer to example 4 and section 7.3 to solve these problems

Check Your Progress 3

Problem-1. Differentiate between Joint, Marginal and conditional probability with an example of each.

Solution - *Please refer to section 7.9 and example 10 to answer these problems.

Problem-2. Explain Dempster Shafer theory with a suitable example.

Solution - *Please refer to section 7.9 and example 10 to answer these problems.

Problem-3. What are different type of evidences? Give suitable example of each.

Solution - *Please refer to section 7.9 and example 10 to answer these problems.

7.12 FURTHER READINGS

1. David Barber ,”Bayesian Reasoning And Machine Learning”, Cambridge University Press
2. John J. Craig, “Introduction to Robotics”, Addison Wesley publication
3. Ela Kumar, “ Artificial Intelligence”, IK International Publications
4. Ela Kumar, “ Knowledge Engineering ”, IK International Publications

